SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005
& ANSI/NCSL Z540-1-1994

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CALIBRATION

Valid To: August 31, 2018 Certificate Number: 1509.02

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations[1]:

I. Dimensional

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Calipers[3,5]	Up to 24 in 24 to 48 in 48 to 96 in	0.6*R* 1100 µin 1500 µin	Gage blocks (field) Gage blocks (lab)
Dial and Test Indicators[3,5]	Up to 3 in	99 µin	Gage blocks, ULM, SuperMic™
Height Gages[3,5]	Up to 36 in	0.6*R*	Gage blocks
Length Standards	Up to 22 in	(37 +1.5*L*) µin	ULM/SuperMic™, gage blocks
Micrometers[3,5]	Up to 4 in (4 to 24) in (24 to 48) in (48 to 96) in	33 µin 0.6*R* 1100 µin 1500 µin	Gage blocks (field) Gage blocks (lab)

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Plain Ring Gages	(0.125 to 12) in	(22 + 1.5*L*) µin	ULM, gage blocks
Thread Plugs-- Pitch Diameter Major Diameter Pitch Diameter (Taper)	 (0.040 to 12) in (0.040 to 12) in (0.040 to 12) in	 (69 + 1.5*L*) µin (18 + 1.5*L*) µin (75+ 1.5*L*) µin	Three-wire method, direct measure Two-wire method, direct measure
Surface Plates[3,5]	(1.5 x 1.5) to (4 x 8) ft (1.5 x 1.5) to (4 x 8) ft	(48 + 0.41*DL*) µin 31 µin	Leveling system (*DL* is diagonal of the plate) Repeat reading gage
Optical Comparators[3,5] X-Axis Y-Axis Angle	 Up to 12 in Up to 12 in (0 to 90) deg	 (250 + 91*L*) µin (250 + 91*L*) µin 0.029 Degrees	 Glass scale Angle blocks
Rulers[3,5] Tape Measures[3,5] PI Tapes[3,5]	Up to 40 in Up to 40 in Up to 100 ft Up to 72 in	0.013 in 0.013 in 0.062 in 910 µin	Standard ruler Standard ruler Standard tape Gage blocks

II. Electrical/DC-Low Frequency

Parameter/Range	Frequency	CMC[2, 7, 8] (±)	Comments
AC Current – Measure 1 A 3 A	 60 Hz to 1 kHz 60 Hz to 1 kHz	 0.1 % + 0.4 mA 0.1 % + 1.8 mA	 HP 34401

Parameter/Range	Frequency	CMC[2, 6, 7, 8] (±)	Comments
AC Current – Generate			
(29 to 330) µA	45 Hz to 1 kHz	0.13 % + 0.1 µA	Fluke 5522A
(330 to 3.3) mA	45 Hz to 1 kHz	0.10 % + 0.15 µA	
(3.3 to 33) mA	45 Hz to 1 kHz	0.07 % + 2 µA	
(33 to 330) mA	45 Hz to 1 kHz	0.083 % + 20 µA	
(0.33 to 3) A	45 Hz to 1 kHz	0.10 % + 100 µA	
(3 to 10) A	45 Hz to 1 kHz	0.21 % + 2 mA	
(16 to 150) A	(60 to 100) Hz	0.21 % + 10 mA	Using 50 turn coil
(150 to 1000) A	(60 to 100) Hz	0.23 % + 0.1 A	
AC Voltage – Generate			
(1 to 330) mV	45 Hz to 10 kHz	0.018 % + 8 µV	Fluke 5522A
(0.330 to 3.3) V	45 Hz to 10 kHz	0.018 % + 60 µV	
(3.3 to 33) V	45 Hz to 10 kHz	0.017 % + 0.6 mV	
(33 to 330) V	45 Hz to 1 kHz	0.023 % + 2 mV	
	(1 to 10) kHz	0.023 % + 6 mV	
(330 to 1020) V	45 Hz to 1 kHz	0.046 % + 10 mV	
	(1 to 5) kHz	0.040 % + 10 mV	
AC Voltage – Measure			
100 mV	60 Hz to 3 kHz	0.069 % + 40 µV	HP 34401
1 V	60 Hz to 3 kHz	0.064 % + 300 µV	
10 V	60 Hz to 3 kHz	0.064 % + 3 mV	
100 V	60 Hz to 3 kHz	0.064 % + 30 mV	
750 V	60 Hz to 3 kHz	0.064 % + 225 mV	

Parameter/Equipment	Range	CMC[2, 6, 8] (±)	Comments
Capacitance – Generate			
Low	(0.22 to 3.3) nF	0.5 % + 10 pF	Fluke 5522A
	(3.3 to 33) nF	0.26 % + 100 pF	
	(33 to 330) nF	0.26 % + 300 pF	
	(330 to 3300) nF	0.26 % + 3 nF	

Parameter/Equipment	Range	CMC[2, 6, 7, 8] (±)	Comments
Capacitance – Generate (cont)			
High	(3.3 to 33) µF (33 to 330) µF (330 to 3300) µF (3.3 to 110) mF	0.4 % + 30 nF 0.45 % + 300 nF 0.45 % + 3 µF 1.1 % + 10 µF	Fluke 5522A
DC Current - Generate	(0 to 330) µA Up to 3.3 mA Up to 33 mA Up to 330 mA Up to 3 A (1.1 to 11) A (11 to 20.5) A	0.015 % + 20 nA 0.010 % + 50 nA 0.014 % + 250 nA 0.010 % + 2.5 µA 0.038 % + 40 µA 0.050 % + 500 µA 0.1 % + 750 µA	Fluke 5522A
	(0 to 55) A (55 to 150) A (150 to 550) A (550 to 1000) A	0.02 % + 2 mA 0.04 % + 2 mA 0.056 % + 25 mA 0.1 % + 38 mA	Using 50 turn coil
DC Current – Measure, Fixed Points	10 mA 100 mA 1 A 3 A	0.076 % + 6 µA 0.05 % + 5 µA 0.1 % + 100 µA 0.14 % + 600 µA	HP 34401
DC Voltage – Generate	(0 to 330) mV Up to 3.3 V Up to 33 V 30 to 330 V (100 to 1000) V	0.002 % + 1 µV 0.0012 % + 2 µV 0.0013 % + 20 µV 0.0018 % + 0.15 mV 0.0019 % + 1.5 mV	Fluke 5522A
DC Voltage – Measure, Fixed Points	100 mV 1 V 10 V 100 V 1000 V	0.0054 % + 3.5 µV 0.0043 % + 7.0 µV 0.0036 % + 50 µV 0.0048 % + 600 µV 0.0047 % + 10 mV	HP 34401

Parameter/Equipment	Range	CMC[2, 6, 7, 8] (±)	Comments
Electical Calibration of Temperature Controllers[3,5]	(-200 to 1371) °C	1.0 °C	Fluke 724 – J, K, T, E
Resistance – Generate	(0 to 330) Ω (330 to 3300) Ω (3300 to 33 000) Ω (33 to 330) kΩ (330 to 3300) kΩ (3.3 to 33) MΩ (33 to 110) MΩ (110 to 330) MΩ (330 to 1100) MΩ	0.065 % + 20 mΩ 0.029 % + 100 mΩ 0.029 % + 1.0 Ω 0.0037 % + 10 Ω 0.0063 % + 150 Ω 0.026 % + 2500 Ω 0.063 % + 3.0 kΩ 0.14 % + 100 kΩ 0.14 % + 500 kΩ	Fluke 5522A
Resistance – Measure, Fixed Points	100 Ω 1 kΩ 10 kΩ 100 kΩ 1 MΩ 10 MΩ 100 MΩ	0.01 % + 4 mΩ 0.01 % + 10 mΩ 0.01 % + 100 mΩ 0.01 % + 1.0 Ω 0.01 % + 10 Ω 0.04 % + 100 Ω 0.8 % + 10 kΩ	HP 34401
RTD Simulation – PT385, 100 Ω	(-200 to 0) °C (0 to 100) °C (100 to 400) °C (400 to 630) °C (630 to 800) °C	0.058 °C 0.17 °C 0.12 °C 0.14 °C 0.28 °C	Fluke 5522A
Thermocouple Simulation – Type E	(-250 to -100) °C (-100 to -25) °C (-25 to 350) °C (350 to 650) °C (650 to 1000) °C	0.51 °C 0.16 °C 0.19 °C 0.20 °C 0.24 °C	Fluke 5522A

Parameter/Equipment	Range	CMC[2, 6] (±)	Comments
Thermocouple Simulation – (cont)			Fluke 5522A
Type J	(-210 to -100) °C (-100 to -30) °C (-30 to 150) °C (150 to 760) °C (760 to 1200) °C	0.28 °C 0.16 °C 0.14 °C 0.17 °C 0.23 °C	
Type K	(-200 to -100) °C (-100 to -25) °C (-25 to 120) °C (120 to 1000) °C (1000 to 1372) °C	0.33 °C 0.18 °C 0.20 °C 0.24 °C 0.40 °C	
Type T	(-250 to -150) °C (-150 to 0.0) °C (0.0 to 120) °C (120 to 400) °C	0.63 °C 0.24 °C 0.16 °C 0.14 °C	
Welding Devices[3,5]	(0 to 350) ADC (0 to 100) VDC (100 to 700 Feed Rate IPM)	0.87 ADC 0.012 VDC 3.6 IPM	Load bank, current shunt, DMM

III. Mechanical

Parameter/Equipment	Range	CMC[2] (±)	Comments
Balances[3, 5]	(0 to 20) g (0 to 200) g	0.11 mg 0.69 mg	Handbook 44 with: Class 1 weights
	(0 to 200) g (0 to 1000) g (0 to 5000) g	1.3 mg 6.6 mg 34 mg	Class 2 weights

Parameter/Equipment	Range	CMC[2] (±)	Comments
Balances (cont)[3, 5]	(0 to 20) g (0 to 200) g (0 to 1000) g (0 to 5000) g (0 to 10 000) g (0 to 20 000) g	0.81 mg 4.8 mg 24 mg 130 mg 370 mg 460 mg	Handbook 44 with: Class 4 weights
Force – Universal Testing Machines[3, 5] –			
Tension / Compression	(0 to 100) lbf	0.0064 lbf	Class F weights w/ load cells
Compression	(100 to 1000) lbf (1000 to 10 000) lbf (10 000 to 50 000) lbf (10 000 to 100 000) lbf (50 000 to 500 000) lbf	0.26 % Indication 0.24 % Indication 0.33 % Indication 0.21 % Indication 0.21 % Indication	
Tension	(100 to 1000) lbf (1000 to 10 000) lbf (10 000 to 50 000) lbf	0.23 % Indication 0.23 % Indication 0.29 % Indication	
Pipettes[3, 5]	(1 to 50) µL (50 to 100) µL (100 to 500) µL (500 to 1000) µL (1000 to 5000) µL (5000 to 10 000) µL	0.31 µL 0.6 µL 1.3 µL 2.2 µL 12 µL 18 µL	Gravimetric method
Optical Tachometer[3, 5]	200 FPM 3000 FPM 29 999 FPM	1.3 FPM 1.7 FPM 3.9 FPM	Calibrated strobe
Pressure Gages, Transducers and Transmitters[3,5]	(0 to 10000) psig (0 to 300) psig	6.6 psig 0.25 psig	Druck DPI 104 gauges
	(0 to 1000) psig (0 to 300) psig (0 to 50) psig (0 to 5) psig (0 to 10) inH20	2.5 psig 0.090 psig 0.019 psig 0.012 psig 0.034 inH20	Druck DPI 620

Parameter/Equipment	Range	CMC[2] (±)	Comments
Vacuum[3,5]	(0 to 15) psig	0.19 psig	Druck DPI 610
Scales[3, 5]	(0 to 10) lb (0 to 20) lb (0 to 50) lb (0 to 100) lb (0 to 200) lb (0 to 500) lb (0 to 1000) lb (0 to 5000) lb (0 to 20 000) lb	0.00025 lb 0.00025 lb 0.0063 lb 0.0063 lb 0.0064 lb 0.0064 lb 0.18 lb 0.19 lb 0.21 lb	Handbook 44 with: Class 4 weights Class F weights
Torque Wrenches[3,5]	(1 to 50) in·lbf (30 to 400) in·lbf (80 to 1000) in·lbf (240 to 3000) in·lbf	0.17 in·lbf 1.8 in·lbf 5.1 in·lbf 21 in·lbf	CDI/Norbar torque calibrator
Rotational Speed Measurement[3,5]	Contact: 200 RPM 3000 RPM 20 000 RPM Optical: 3000 RPM 30 000 RPM	 1.5 RPM 2.4 RPM 11 RPM 1.3 RPM 3.3 RPM	Monarch PLT200
Hardness			
HRB	1 to 50 51 to 79 ≥ 80	1.0 HRB 0.65 HRB 0.49 HRB	Comparison to test blocks
HRC	20 to 39 40 to 59 60 to 70	0.45 HRC 0.36 HRC 0.37 HRC	
HBW 10/500	90 to 200	4.0 HBW	
HBW 10/3000	200 to 600	4.8 HBW	

IV. Thermodynamics

Parameter/Equipment	Range	CMC[2] (±)	Comments
Temperature – Measuring Equipment[3,5]	(30 to 600) °C (86 to 1112) °F	0.35 °C 0.63 °F	Fluke 9144 in dry block
Temperature – Measuring Equipment	(-40 to 90) °C (-40 to 194) °F	0.35 °C 0.63 °F	Fluke 9144 probe in wet bath (lab only)
Temperature – Measure[3,5]	(-100 to 600) °C (-148 to 1112) °F	0.81 °C 1.46 °F	Fluke 724 with thermocouple
IR Thermometer[3,5]	(Amb +10 to 400) °C	1.8 °C	Omega BB703
Humidity – Measure[3,5]	(10 to 90) %RH	3.0 % RH	Vaisala HM141/HM046

V. Time & Frequency

Parameter/Equipment	Range	CMC[2] (±)	Comments
Frequency – Measure	40 Hz to 300 kHz	0.014 % Indication	HP 34401
Timers and Stopwatches[3, 5]	< 1 hr ≥ 1 hr	0.14 sec/24 hr 0.59 sec/24 hr	Reference timer

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[1] This laboratory offers commercial calibration service and field calibration service.

[2] Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of k = 2. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

[3] Field calibration service is available for this calibration and this laboratory meets A2LA *R104 – General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution, repeatability) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

[4] In the statement of CMC, *L* is the numerical value of the nominal length of the device measured in inches and *R* is the numerical value of the resolution of the device in microinches unless otherwise noted.

[5] The CMC stated for calibrations performed in the laboratory is applicable for calibrations performed in the field.

[6] CMC for the Fluke 5522A is based on 1-year specifications within a temperature range of 23 °C ± 5 °C. Field calibrations will be performed within 23 °C ± 5 °C .

[7] CMC for the HP 34401 is based on 1-year specifications within a temperature range of 18 °C to 28 °C. Field calibrations will be performed within 18 °C to 28 °C, 30 % to 55 % humidity.

[8] The stated measured values are determined using the indicated instrument (see Comments). This capability is suitable for the calibration of the devices intended to measure or generate the measured value in the ranges indicated. CMC's are expressed as either a specific value that covers the full range or as a percent or fraction of the reading plus a fixed floor specification.

Accredited Laboratory

A2LA has accredited

ALDINGER COMPANY

Little Rock, AR

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSLI Z540-1-1994 and any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (*refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009*).

AMERICAN ASSOCIATION FOR LABORATORY ACCREDITATION
CORPORATE SEAL 1978
DISTRICT OF COLUMBIA
A2LA

Presented this 14th day of June 2016.

Senior Director of Quality and Communications
For the Accreditation Council
Certificate Number 1509.02
Valid to August 31, 2018

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.